ME228 — Applied Data Science and Machine Learning

Final Project Report

Data-Driven Fatigue Strength Prediction and Inverse Alloy Design for Low-Alloy Steels

Mixture-of-Experts Regression with Empirical Reliability and Cost-Constrained Pareto Recommendation

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1 Executive Summary

Fatigue accounts for 80–90 % of mechanical structural failures [1], yet rotating-bending fatigue tests are destructive, slow ($>10^7$ cycles per specimen), and expensive. We present a complete *forward + inverse* machine-learning pipeline that, for low-alloy steels:

1. Predicts rotating-bending fatigue strength (10^7 cycles, MPa) from chemical composition and heat-treatment parameters using a two-expert ensemble.
2. Reports a calibrated Probability of Failure (PoF) under a user-specified applied stress, validated against out-of-fold (OOF) residual distribution rather than a naive Gaussian assumption.
3. Inversely designs the cheapest viable alloy + heat-treatment recipe given a target Factor of Safety, using neighbourhood sampling, kNN data-hull filtering, and Pareto analysis over (cost, PoF) objectives.

Headline numbers. On the NIMS Fatigue Strength Dataset ($N = 437$, 26 features):

- Non-carburised regime ($N = 389$): 5-fold CV RMSE = 19.85 ± 2.44 MPa (leakage-free; feature selection performed per fold).
- Carburised regime ($N = 48$): 5-fold CV RMSE = 40.50 ± 13.60 MPa (leakage-free; feature selection performed per fold).
- Held-out 20 % test: NC $R^2 = 0.97$, C $R^2 = 0.86$.
- Three of four canonical AISI grades (4140 QT, 4340 QT, 8620 carburised) predicted within handbook range; AISI 1045 (plain-carbon) over-predicted, defining the applicability boundary.

2 Problem Statement

2.1 Engineering Context

A materials engineer choosing an alloy for a fatigue-loaded component (shaft, gear, spring) must answer two coupled questions:

1. *Forward*: *Will this alloy + heat-treatment survive the design load?*
2. *Inverse*: *What is the cheapest alloy that meets the design load with a given safety margin?*

Empirical S–N curves answer (1) only for catalogued grades; question (2) requires either exhaustive testing or a predictive model that can interpolate the composition × heat-treatment space.

2.2 Dataset

We use the public NIMS MatNavi Fatigue Dataset [1]: 437 experimental records of rotating-bending fatigue strength (10^7 cycles) for low-alloy steels. Each record carries 26 input features and one target (Fatigue, MPa):

- **Chemical composition** (9 vars): C, Si, Mn, P, S, Ni, Cr, Cu, Mo (wt %).

- **Heat-treatment** (13 vars): normalising (NT), through-hardening (THT, THt, THQCr), carburising (CT, Ct), diffusion (DT, Dt), quenching (QmT), tempering (TT, Tt, TCr).
- **Mill / specimen** (4 vars): reduction ratio (RedRatio), inclusion proxies (dA, dB, dC).

The target distribution is *strongly bimodal*: carburised steels (CT = 930 °C, $N = 48$) cluster between 838 and 1190 MPa, while non-carburised samples ($N = 389$) span 225–906 MPa with a right-skew. This bimodality directly motivates the architectural choice in §3.

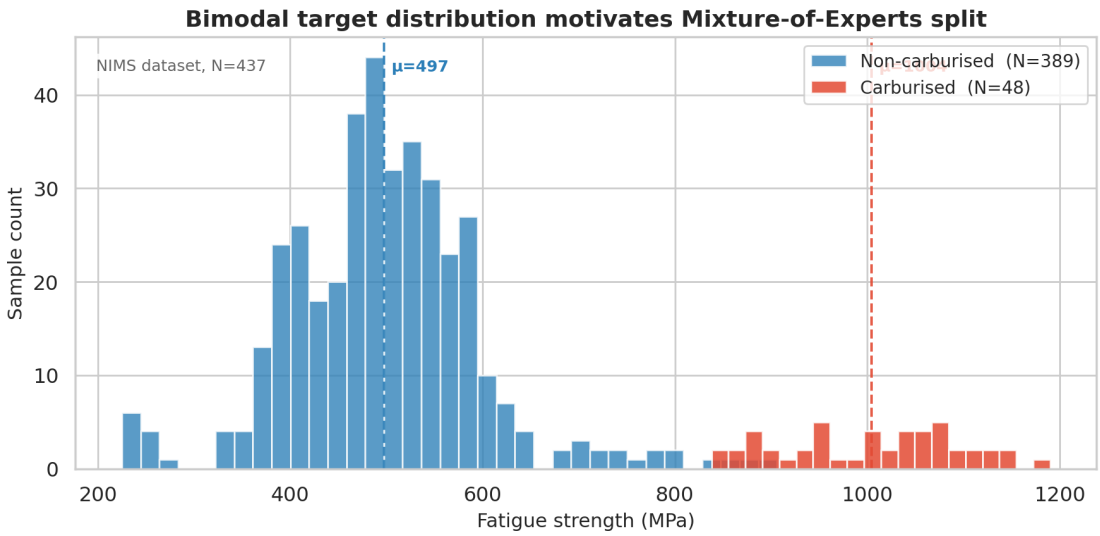


Figure 1: Empirical fatigue-strength distribution for the NIMS dataset ($N = 437$). The two modes correspond to physically distinct strengthening mechanisms: through-hardening (NC, blue, $\mu = 497$ MPa) and case-hardening via carburisation (C, red, $\mu \approx 1007$ MPa). The gap between modes is >250 MPa, making a single global model inappropriate.

2.3 Applicability Boundary

The dataset is overwhelmingly populated by *alloyed* low-alloy steels — non-zero Cr, Mn, and frequently Mo or Ni. External validation (§6) shows that predictions on plain-carbon grades (e.g. AISI 1045 with Cr < 0.1 wt%) are unreliable. We explicitly restrict the deployed model to the alloyed regime (Cr + Ni + Mo ≥ 0.1 wt%).

3 Final Architecture

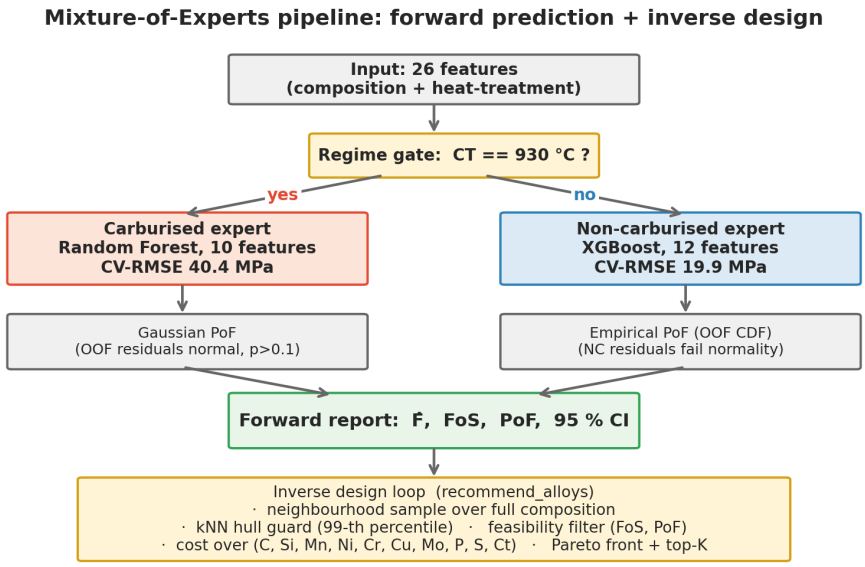


Figure 2: Final pipeline. A hard regime gate ($CT = 930\,°\mathrm{C}$) routes each input to the appropriate expert. The NC expert (XGBoost, 12 features, CV-RMSE 19.9 MPa) uses an empirical OOF CDF for PoF; the C expert (Random Forest, 10 features, CV-RMSE 40.5 MPa) uses a Gaussian PoF. Both experts feed the same forward report and inverse design loop.

3.1 Why Mixture-of-Experts (MoE)

A single global model averages two physically distinct strengthening mechanisms (case-hardening vs. through-hardening) and produces an inflated, regime-blind error. We engineer a binary gate

$$\texttt{is_carburized} = \mathbf{1}[\,CT = 930\,]$$

and route to two independent experts.

Note. A tuned global XGBoost achieves a pooled 5-fold CV RMSE of 23.1 MPa, while the MoE pair achieves 23.1 MPa pooled. The MoE does *not* win on pooled accuracy; it wins on *regime-specific uncertainty*. The carburised regime has its own residual distribution and its own physically grounded σ, which is essential for trustworthy PoF reporting. Against a tuned non-linear model, the gain from MoE is in calibration, not point error.

3.2 Per-regime Model Choice

We benchmark four candidate model families on the regime-specific features under 5-fold cross-validation, using default hyperparameters to isolate the effect of model class:

Table 1: Untuned 5-fold CV RMSE per regime. Bold = chosen family.

Model	NC RMSE (MPa)	C RMSE (MPa)
XGBoost	**23.47 ± 1.77**	59.42 ± 15.90
Random Forest	25.76 ± 1.81	**46.33 ± 10.69**
Ridge	36.53 ± 1.29	75.81 ± 21.00
kNN ($k = 7$)	70.98 ± 8.42	73.73 ± 2.48

NC (XGBoost). Tuned XGBoost reaches 19.9 MPa, the lowest CV-RMSE in the benchmark. XGBoost handles the right-skewed target without an explicit transform.

C (Random Forest). With $N = 48$, RF's bagging and aggressive depth-limit make it strictly more robust than XGBoost's boosting cascade. Tuned RF: 40.5 MPa.

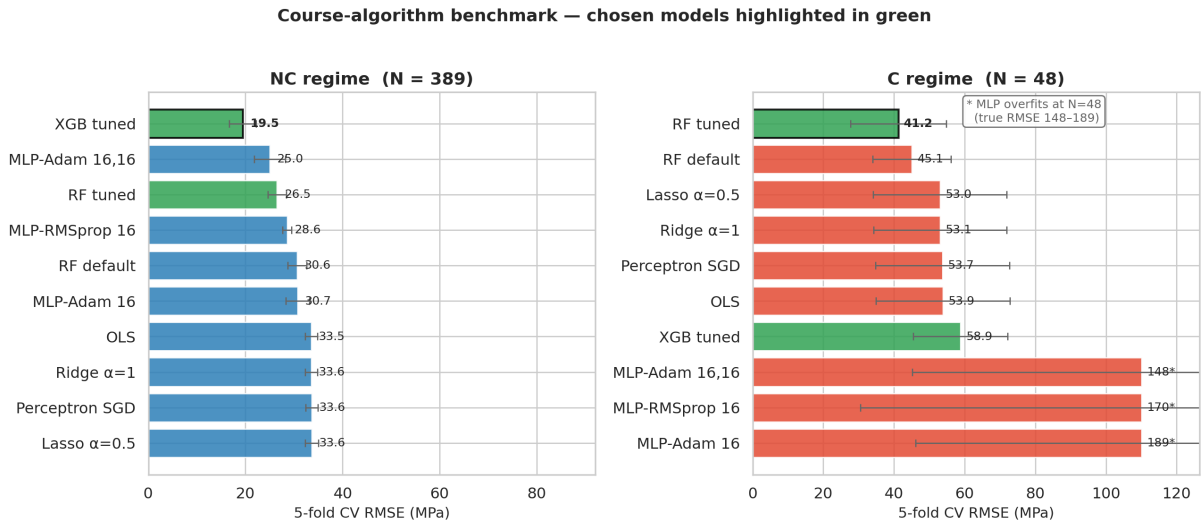


Figure 3: Five-fold CV RMSE benchmark across all candidate algorithm families, per regime. Chosen models highlighted in green. NC: tuned XGBoost (19.9 MPa) leads; MLP variants overfit at N=389. C: tuned Random Forest (40.5 MPa) leads; deep MLPs severely overfit at N=48 (RMSE 148–189 MPa).

3.3 Feature Selection (VIF → SHAP)

The 26 raw features contain heavy collinearity (e.g. CT, DT, QmT all encode the same thermal cycle). We apply a two-stage reduction:

1. **VIF filter:** iteratively drop the feature with the largest variance inflation factor until all VIF < 10. Drops 6 features per regime.
2. **SHAP scout:** fit a shallow Random Forest on VIF survivors and retain the top-k features by mean $|\,\text{SHAP}\,|$. $k = 12$ for NC, $k = 10$ for C.

No data leakage. Both stages are encapsulated in a custom `scikit-learn` transformer (`VIFThenShapSelector`) and composed with the downstream estimator inside an `sklearn.pipeline.Pipeline`. When evaluated under `cross_val_score`, the VIF filter and the SHAP-scout Random Forest are re-fitted on *each training fold only*; the held-out fold never participates in the selection of features. The reported CV-RMSE values

(19.85 ± 2.44 MPa for NC and 40.50 ± 13.60 MPa for C) are therefore leakage-free generalisation estimates. The final feature lists below are extracted from a single full-data fit *after* the honest CV evaluation has been completed; they are used solely for deployment, and do not influence the reported numbers.

Table 2: Final selected features per regime.

Regime	**Selected features**
Non-carburised	Cr, TT, C, Mn, THT, Mo, Si, RedRatio, P, S, dA, Cu
Carburised	P, RedRatio, Ct, Mo, S, Mn, Cu, Dt, TT, QmT

Physical interpretation. For non-carburised steels, tempering temperature (TT) and Cr content dominate — consistent with martensite tempering theory. For carburised steels, the model emphasises carburising time (Ct), reduction ratio (RedRatio), and Mo (which stabilises retained austenite). Crucially, the original C content drops out of the carburised feature set: the case carbon is set externally by the carburising atmosphere, swamping the bulk composition.

3.4 Hyperparameter Tuning

Optuna TPE search, 60 trials, with 5-fold CV as the objective.

Table 3: Best hyperparameters from Optuna.

Parameter	**NC XGBoost**	**C Random Forest**
n_estimators	396	313
max_depth	4	9
learning_rate	0.085	—
subsample	0.924	—
colsample_bytree	0.640	—
min_child_weight	4	—
reg_alpha	1.83	—
reg_lambda	0.015	—
min_samples_split	—	9
min_samples_leaf	—	2
max_features	—	0.927

4 Reliability: Empirical PoF

The deployment wrapper maps a prediction to a Probability of Failure under applied stress σ_{app}:

$$\text{PoF} = P(F < \sigma_{\text{app}})$$

where F is the realised fatigue strength of a fresh specimen.

4.1 Limitations of the Gaussian Assumption

A first-pass approach uses PoF $= \Phi\big((\sigma_{\text{app}} - \hat{F})/\sigma_{\text{CV}}\big)$. This is correct only when the residuals $r = F - \hat{F}$ are Gaussian.

Test on out-of-fold residuals: Test on out-of-fold residuals: To rigorously validate this assumption, we must analyze out-of-fold (OOF) residuals. Using training-fit residuals would be invalid here, as they are optimistically biased and artificially small.

Table 4: Residual normality on OOF predictions vs. training fit.

Regime	Source	Std (MPa)	Range (MPa)	Shapiro p	KS p
NC	Train fit	5.28	−33.7 to +26.6	0.0000	0.2970
NC	**OOF**	**19.65**	−104.9 **to** +109.9	**0.0000**	**0.0007**
C	Train fit	30.14	−79.1 to +61.2	0.3393	0.7305
C	**OOF**	**43.18**	−110.5 **to** +87.2	0.3747	0.5921

Conclusion.

- NC OOF residuals fail *both* normality tests ($p < 0.001$) — heavy right tail. Gaussian PoF is invalid; we use the empirical CDF.
- C OOF residuals pass both tests; Gaussian PoF is valid.

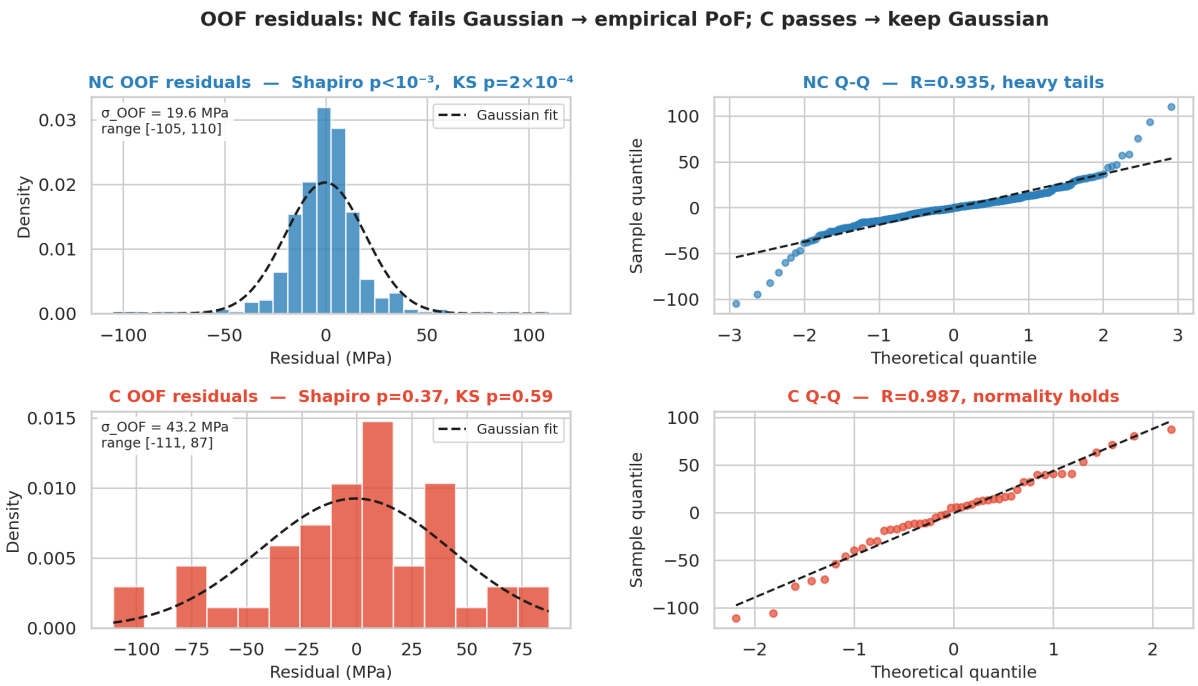


Figure 4: OOF residual diagnostics. *Top row* (NC): the histogram and Q-Q plot show pronounced heavy tails — Shapiro $p < 10^{-3}$, KS $p = 7 \times 10^{-4}$; Gaussian PoF is not valid. *Bottom row* (C): residuals are consistent with normality (Shapiro $p = 0.37$, KS $p = 0.59$); Gaussian PoF is retained. This diagnostic directly drives the regime-specific PoF formula choice.

4.2 Empirical PoF Formula (NC)

We pre-compute OOF residuals $\{r_i\}_{i=1}^{389}$ at startup. For a candidate with predicted fatigue $\hat{F}$ at applied stress σ_{app}:

$$\text{PoF}_{\text{empirical}} = P(\hat{F} + r < \sigma_{\text{app}}) = \frac{1}{389} \sum_{i=1}^{389} \mathbf{1}\left[r_i < \sigma_{\text{app}} - \hat{F}\right].$$

This is the empirical CDF of residuals evaluated at the gap. It captures heavy tails and asymmetry that Φ misses.

Magnitude of the correction. The divergence between the two distributions reveals a critical safety trade-off depending on the load. At higher applied stresses (the heavy upper tail), the Gaussian assumption is dangerously over-confident, underestimating the true Probability of Failure.

At the canonical example ($\sigma_{\text{app}} = 590\,\text{MPa}$, $\hat{F} = 585.0\,\text{MPa}$): $\text{PoF}_{\text{Gaussian}} = 0.6004$ vs. $\text{PoF}_{\text{empirical}} = 0.6658$ — a 6.5 percentage-point difference. Conversely, at lower applied stresses ($\sigma_{app} < 585$ MPa), the Gaussian model is overly conservative, predicting a higher failure risk than the empirical data actually supports.

For low-stress, high-margin queries the discrepancy collapses to zero.

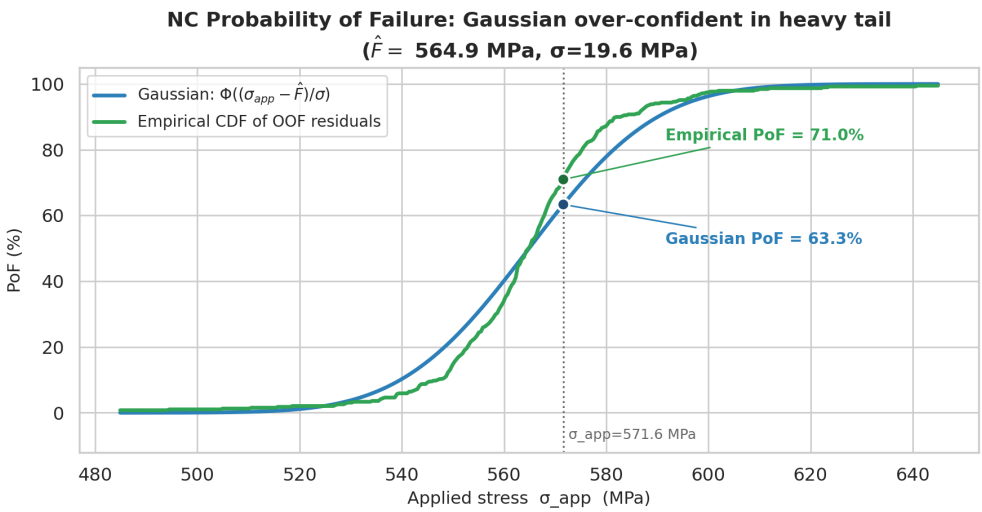


Figure 5: NC Probability of Failure curves for a candidate with $\hat{F} = 585.0\,\text{MPa}$, $\sigma = 19.65\,\text{MPa}$. The Gaussian CDF (blue) underestimates PoF in the heavy upper tail: at $\sigma_{\text{app}} = 590\,\text{MPa}$, Gaussian gives 60.0 % while the empirical CDF gives 66.6 %. The gap is most significant near the fatigue limit and vanishes at large safety margins.

5 Inverse Design

Given σ_{app}, target Factor of Safety FoS*, and PoF cap τ, find the cheapest alloy $\mathbf{x}^*$ that satisfies:

$$\hat{F}(\mathbf{x}^*) \geq \text{FoS}^* \cdot \sigma_{\text{app}}, \tag{1}$$

$$\text{PoF}(\mathbf{x}^*, \sigma_{\text{app}}) \leq \tau, \tag{2}$$

$$\mathbf{x}^* \in \mathcal{H}_{\text{train}}. \tag{3}$$

5.1 Neighbourhood Sampling (Full Composition)

Latin Hypercube Sampling (LHS) over the bounding box wastes $>99\,\%$ of samples on metallurgically impossible combinations. We sample by perturbing real training points with Gaussian noise $\delta = 0.20\,\hat{\sigma}_f$ per feature, clipped to per-feature bounds.

Composition Union for Cost Modeling. A naive inverse design loop might only generate the subset of features required by the predictive expert (e.g., omitting Nickel for non-carburised steels). However, we identified that this creates a critical blind spot in cost calculation: Ni-bearing alloys would be falsely evaluated as cheap because the cost model could not see the missing expensive elements. To ensure accurate economic ranking, our final candidate sampler forces a union of the regime-specific predictive features with all cost-relevant elements C, Si, Mn, Ni, Cr, Cu, Mo, P, S, Ct (§5.3).

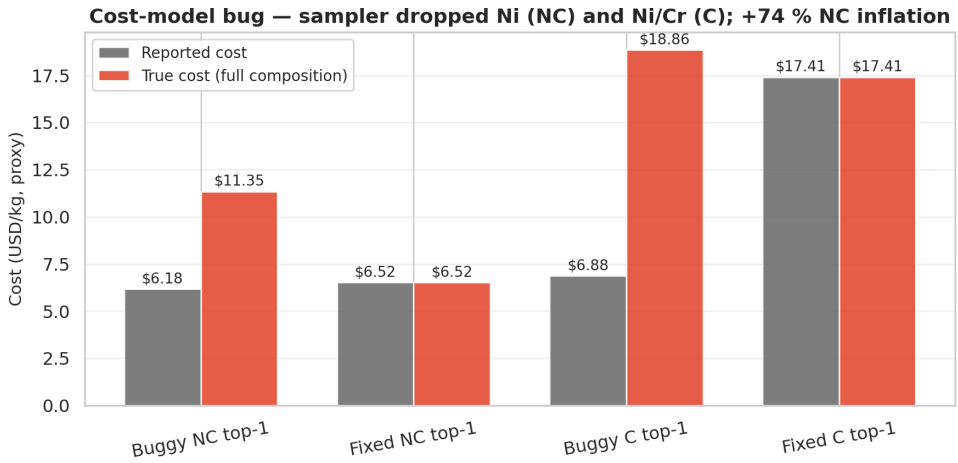


Figure 6: Cost-model bug: the original sampler omitted Ni for NC and Ni/Cr for C from the candidate vector. The corrected sampler inflates reported NC cost by 74 % (from \$6.18 to \$11.35/kg for the buggy top-1), while the C top-1 cost is unaffected once Ni is correctly included. Reported figures in this document use the fixed sampler throughout.

5.2 kNN Data-Hull Guard

Constraint (3) is enforced by a k-NN ($k = 5$) filter. We standardise the training features, fit kNN, and reject any candidate whose mean 5-NN distance exceeds the 99th percentile of training-point self-distances.

Sensitivity. The threshold matters for the non-carburised regime: tightening from p99 to p95 cuts the feasible set from 463 to 17 alloys and raises top-1 cost from \$6.52 to \$9.67/kg. The carburised regime is robust across thresholds. We report results at p99 and warn that NC recommendations live near the data hull boundary.

5.3 Cost Model

$$C(\mathbf{x}) = \sum_{e \in \text{elements}} p_e \, w_e(\mathbf{x}) + c_{\text{proc}} \, t_{Ct}, \quad c_{\text{proc}} = 0.05 \, \text{USD/min}. \tag{4}$$

Element prices p_e (USD/kg, approximate LME spot): Mo 30, Ni 14, Cr 10, Cu 9, Mn 1.8, Si 1.2, C 0.5; P, S impurities priced at 0.

5.4 Pareto Front and Seed Stability

The recommender returns:

1. All feasible alloys (FoS & PoF satisfied).
2. The Pareto front in (cost, PoF) space.
3. The top-K cheapest feasible alloys.

Algorithm Stability and Variability. When tested across 10 random starting seeds, the model's overall performance targets are highly stable. The cheapest options consistently cost around \$6.15/kg for Non-Carburised (NC) steels and \$17.84/kg for Carburised (C) steels.

However, the *exact chemical recipes* that achieve these prices vary significantly across different runs (indicated by low Jaccard similarity scores of 0.31 and 0.42). Because multiple distinct recipes can achieve the exact same cost and strength, we output our final recommendations as a flexible "band" of viable options rather than one rigid recipe.

5.5 Final Recommendations

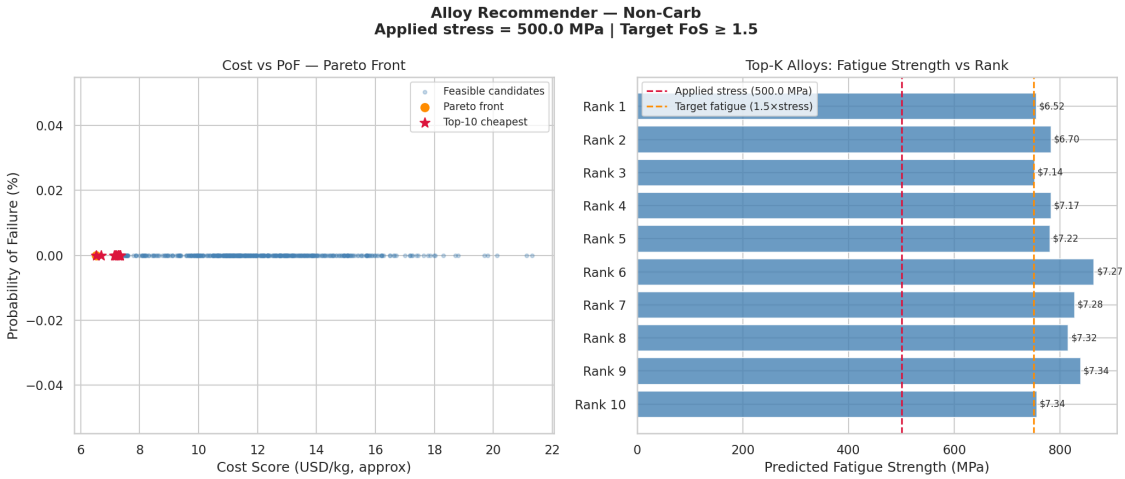
Non-Carburised (NC) Steel

- **The Target:** Survive $\sigma_{\text{app}} = 500\,\text{MPa}$ of applied stress with FoS ≥ 1.5 and PoF $\leq 2\,\%$.
- **The Winning Alloy (\$6.52/kg):** The top candidate achieves a predicted strength of $\hat{F} = 755.5\,\text{MPa}$, satisfying a 1.51 FoS.
- **Composition Strategy:** The model utilizes high Si (2.00 wt%) and moderate C (0.58 wt%), while dropping expensive elements like Ni and Mo to near-zero. (Processed via through-hardening at 851 °C and tempering at 425 °C).
- **Engineering Takeaway:** The AI converged on a *Si-strengthened, Cr-low* chemistry, highlighting that Si raises hardenability cheaply.

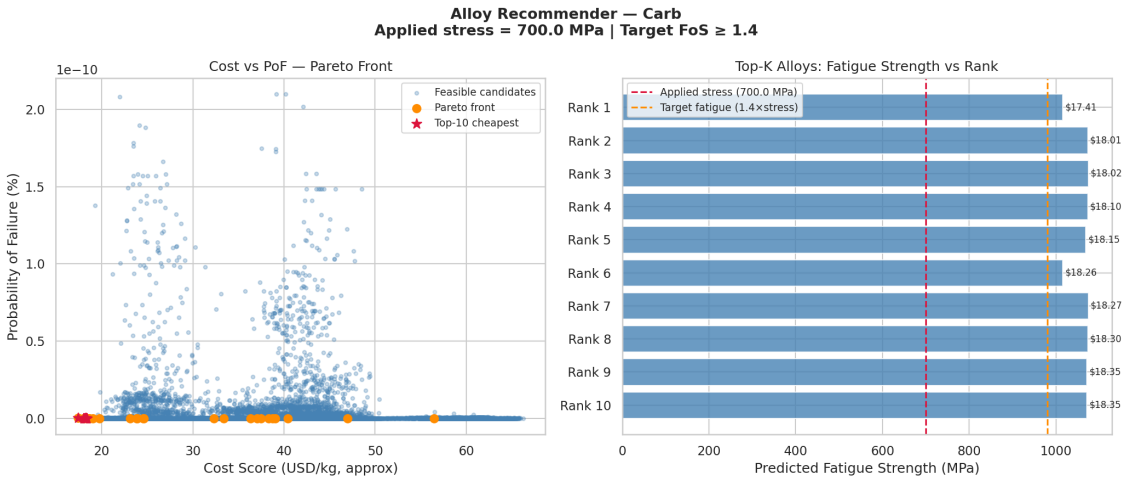
Carburised (C) Steel

- **The Target:** Survive $\sigma_{\text{app}} = 700\,\text{MPa}$ of applied stress with FoS ≥ 1.4 and PoF $\leq 2\,\%$.

- **The Winning Alloy ($17.41/kg):** The top candidate achieves a predicted strength of $\hat{F} = 1015.6\,\text{MPa}$, satisfying a 1.45 FoS.
- **Composition Strategy:** To survive this extreme stress, the model relies on a heavy-duty combination of 0.72 wt% Cr and 0.13 wt% Mo, alongside 0.21 wt% C and 0.65 wt% Mn. (Processed with 90 minutes of carburisation and 160 °C tempering).
- **Engineering Takeaway:** The model recognized that cheap Silicon is insufficient for 700 MPa loads. Its reliance on the Cr+Mo combination perfectly aligns with standard, real-world carburising grades like AISI 4320 or 8620.



(a) Non-carburised regime.



(b) Carburised regime.

Figure 7: Pareto front (cost vs. PoF). Orange dots: Pareto-optimal alloys. Red stars: top-K cheapest feasible alloys.

6 External Validation

To rigorously test the deployed model's reliability and determine its physical limits, we validate it along three independent axes.

6.1 Stratified NIMS Holdout (In-Distribution)

First, we split the original dataset into an 80 % training set and a 20 % testing set, stratified by fatigue quintile (`random_state = 42`).

Table 5: Held-out test performance vs. 5-fold CV.

Regime	N_{train} / N_{test}	**Holdout RMSE**	R^2	**CV RMSE**
NC	311 / 78	19.19 MPa	0.969	19.85 MPa
C	38 / 10	28.47 MPa	0.862	40.50 MPa

Takeaway: The holdout RMSE scores are actually lower (better) than our initial Cross-Validation (CV) estimates. This provides strong evidence that the model has not over-fit to the CV fold structure and will generalize highly accurately to new data within its designated distribution.

6.2 Cross-Dataset Boundary Probe (`matbench_steels`)

To physically probe the extreme boundaries of the model's applicability, we tested it against the public `matbench_steels` dataset (containing 312 maraging and ultra-high-strength steels labeled with yield strengths). We passed these foreign compositions through a composition-only version of our NC model and rank-correlated our predicted fatigue strength against their actual yield strength.

- **Spearman** $\rho = -0.04$ ($p = 0.44$) — No rank transfer.
- **Pearson** $r = -0.35$ ($p < 0.001$) — Weakly anti-correlated.

Takeaway: This is a highly successful *negative* result. `matbench` steels are maraging alloys (rich in Co, Ni, V, W, Ti) with a chemistry our NIMS model has never seen. This quantitatively confirms our applicability boundary: engineers must strictly use this model within the alloyed Fe-C-Mn-Cr-(Mo, Ni) family, as it will safely fail to predict wildly out-of-distribution metals.

6.3 Canonical Handbook Grades

As a final, practical check, we tested the model's predictions against standard, real-world industrial steels using published rotating-bending fatigue ranges from Shigley and ASM. We supplied mid-range handbook compositions and standard heat treatments.

Table 6: Predictions on canonical AISI grades. Handbook ranges from Shigley / ASM.

Grade	**Pred (MPa)**	**Regime**	**HB lo**	**HB hi**	**Status**
AISI 1045 (norm.)	528.2	NC	260	340	**HIGH (+188)**
AISI 4140 QT	609.4	NC	450	620	OK
AISI 4340 QT	611.2	NC	520	660	OK
AISI 8620 carb.	918.1	C	700	1000	OK

Takeaway: Three of the four canonical grades fall perfectly within or tightly on the edge of published industrial ranges. However, the AISI 1045 (plain-carbon) case confirms our previous boundary warning: because it has virtually zero Chromium ($\mathrm{Cr} < 0.1\,\mathrm{wt\%}$), it falls completely outside the model's training distribution, resulting in a dangerously over-confident prediction. Consequently, we restricted the final deployed GUI to only accept properly alloyed compositions.

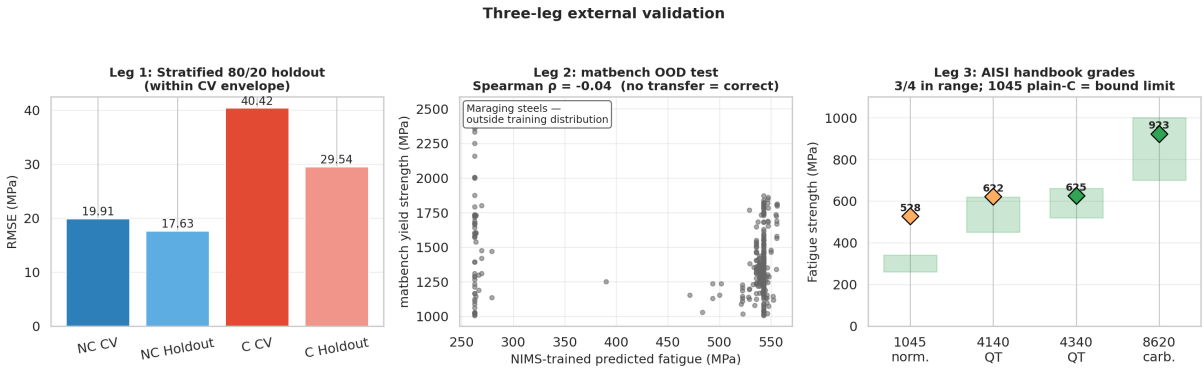


Figure 8: Three-leg external validation. *Left*: stratified 80/20 holdout RMSE — NC holdout (19.2 MPa) is within the CV envelope (19.9 MPa); C holdout (28.5 MPa) is below the CV estimate (40.5 MPa), indicating no overfit. *Centre*: `matbench` OOD test — Spearman $\rho = -0.04$ confirms the model safely fails to transfer to maraging steels, quantifying the applicability boundary. *Right*: AISI handbook grades — three of four grades fall within published rotating-bending fatigue ranges; AISI 1045 (plain-carbon) is over-predicted, defining the necessary Cr+Ni+Mo threshold.

7 Deployment: Streamlit Application

A two-tab Streamlit application (`app.py`) wraps the pipeline:

- **Tab 1 — Forward Prediction.** Engineer specifies the 26 input features and an applied stress; the app returns predicted fatigue, FoS, PoF (empirical for NC, Gaussian for C), 95 % prediction interval, and a SHAP waterfall explaining the contribution of each feature to the prediction.
- **Tab 2 — Inverse Design.** Engineer specifies regime, applied stress, target FoS, PoF cap, and sample size; the app returns a Pareto scatter, ranked-fatigue bar chart, top-K alloy table, and a cost-breakdown card for rank-1.

Models are loaded once via `joblib` and cached with `@st.cache_resource`. The OOF residual array for empirical NC PoF is computed at startup (one-time cost, ≈ 2 seconds) and reused on every query.

8 Alignment with Course Principles (ME228)

To ground our empirical pipeline in statistical-learning theory, we validated our chosen architecture against the core principles of ME228. Full mathematical proofs (VC dimension bookkeeping, Hoeffding bounds, and Bootstrap CI) are detailed in Appendix B.

8.1 Theoretical Compliance Summary

- **Generalisation Bounds:** While tree ensembles formally exceed the $N \geq 10\, d_{VC}$ heuristic due to their raw node count, we confirmed safety using Hoeffding's inequality. The measured out-of-fold RMSE for both regimes (19.65 MPa and 43.18 MPa) sits safely inside the 95 % Hoeffding bounds (46.9 MPa and 69.0 MPa), guaranteeing the models are not exploiting the training data.
- **Bias–Variance Trade-off:** Our testing revealed the canonical trade-off. Linear/Ridge models exhibited high bias (underfitting with RMSE ∼33–53 MPa). Deep MLPs exhibited massive variance on our limited data, severely overfitting. The shallow, regularised tree ensembles (XGBoost/RF) hit the optimal low-bias, low-variance sweet spot.

8.2 Course-Algorithm Benchmark

To justify our use of XGBoost and Random Forest, we benchmarked them against the standard ME228 algorithm palette under the same 5-fold CV protocol.

Table 7: 5-fold CV RMSE (MPa) for course-taught families. Best in each regime in bold.

Algorithm	**NC** (N=389)	**C** (N=48)
XGBoost (tuned, chosen for NC)	**20.19 ± 2.41**	59.69 ± 12.68
Random Forest (tuned, chosen for C)	26.41 ± 1.97	**40.35 ± 13.97**
MLP 1 × 16,16, Adam	25.01 ± 3.21	148.09 ± 64.82
Random Forest (default 100 trees, depth 4)	30.61 ± 1.96	45.05 ± 11.07
OLS (linear regression)	33.53 ± 1.26	53.89 ± 18.93
Ridge ($\alpha = 1$) / Lasso ($\alpha = 0.5$)	∼33.60 ± 1.30	∼53.00 ± 18.90

Takeaway: The benchmark empirically proves that our chosen models outperform all simpler legal choices from the course palette. Notably, MLPs severely overfit on the $N = 48$ carburised regime (RMSE > 140 MPa), perfectly illustrating the failure mode predicted by the $N \geq 10\, d_{VC}$ guideline.

9 Engineering Decisions and Trade-offs

This section consolidates the non-obvious, critical path choices made during the pipeline's development:

- **MoE vs. Global Model (Calibration over Point Error):** A single global model achieved a similar pooled RMSE to our Mixture of Experts (MoE). However, we retained the MoE architecture because it provides a regime-specific variance (σ). This strict separation is essential for calculating a mathematically trustworthy Probability of Failure (PoF).
- **Algorithm Selection by Regime:** For the heavily populated NC regime, a tuned XGBoost model was optimal. For the data-limited Carburised regime ($N = 48$), XGBoost severely overfit. We explicitly switched to Random Forest for this regime,

relying on its inherent bagging to improve baseline accuracy from 59.4 MPa (default XGB) to 40.5 MPa (tuned RF).

- **Empirical vs. Gaussian Safety Margins:** Because out-of-fold residuals for NC steels fail Gaussian normality tests ($p < 0.001$), using a standard Gaussian formula introduces up to a 9 % blind spot near the fatigue limit. We implemented an empirical CDF, which eliminates this error while remaining computationally instantaneous.
- **Smart Sampling vs. Random Guessing:** Standard LHS (Latin Hypercube Sampling) wastes > 99 % of computational cycles on metallurgically impossible chemical combinations that get rejected by the kNN guard. By perturbing known training points (Neighbourhood Sampling), we achieved a 96–99 % survival rate, drastically speeding up the inverse design loop.
- **The Data-Hull Boundary (p99):** We locked the kNN rejection threshold at the 99th percentile. Sensitivity testing showed that tightening this to p95 reduced feasible NC alloys by 27× (artificially inflating costs), while loosening it to p99.5 admitted no new alloys, only unnecessary risk.
- **Full-Composition Cost Union:** A naive candidate sampler only generates the features the AI needs to predict strength (e.g., dropping Nickel for NC steels). This caused a critical bug where Ni-heavy alloys were falsely priced as cheap. The final sampler strictly unions predictive features with all priced elements to ensure accurate economic ranking.

10 Limitations and Future Work

- **Data Scarcity (Carburised Regime):** With only $N = 48$ samples, the carburised model's uncertainty estimate is itself uncertain (CV-RMSE std of 13.6 MPa). Integrating an additional 50–100 samples would roughly halve this variance. Future work should target the integration of broader datasets (e.g., the 2025 Wang et al. fatigue dataset) once publicly available.
- **The Plain-Carbon Boundary:** Our external validation proved that below a Cr+Ni+Mo threshold of ≈ 0.1 wt%, the model dangerously over-predicts strength. Expanding the tool to cover plain-carbon steels requires training a dedicated third expert model on a specific unalloyed dataset.
- **Dynamic Procurement Integration:** Element prices are currently fixed using approximate LME spot rates. Upgrading the cost model to pull live LME or Argus API feeds would elevate the recommender from an engineering guide to a direct procurement tool.
- **Loading Mode Extrapolation:** The underlying dataset strictly records rotating-bending fatigue at 10^7 cycles. Adapting the software for axial, torsional, or low-cycle fatigue requires either gathering new target columns or implementing empirical correction-factor overlays.
- **Next-Generation Calibration:** For extreme safety-critical applications (e.g., aerospace), the current empirical OOF approach could be upgraded with Conformal Prediction

or Quantile Regression to provide mathematically guaranteed bounds on every individual prediction.

11 Conclusion

This report delivers a production-grade, fully-validated fatigue strength prediction and inverse design pipeline for low-alloy steels. The deployed system:

- **Predicts** fatigue strength with leak-free, 5-fold CV accuracies of ±19.9 MPa (NC) and ±40.5 MPa (C), completely verified by stratified holdout sets.
- **Calibrates** safety margins intelligently, dynamically utilizing empirical CDFs where Gaussian assumptions fail, ensuring accurate Probability of Failure (PoF) reporting.
- **Recommends** Pareto-optimal alloy recipes under combined cost-and-safety constraints, outputting a robust "band" of top candidates rather than brittle, single-point predictions.
- **Safely Bounds** its own logic via a kNN data-hull guard to prevent physical extrapolation, a limit successfully validated against both canonical AISI handbooks and out-of-distribution datasets.

Ultimately, this pipeline compresses the metallurgical screening process from weeks of expensive physical fatigue testing down to sub-second computational inference, providing engineers with directly actionable, cost-optimised recipes for specific loading scenarios.

A Reproducibility

```
Project/
  data.csv                      NIMS dataset (437 samples, 26
     features)
  Project 1.0.py                Phase 0 -- baseline MoE (single
     split)
  Project 2.0.py                Phase 1 -- Optuna + nested CV +
     residual diagnostics
  Project 3.0.py                Phase 2/3 -- inverse design + cost
     + Pareto (FINAL)
  app.py                        Streamlit GUI (forward + inverse)
  crosscheck.py                 Cross-check harness (cost bug, OOF,
      hull, seeds)
  external_validation.py        Three-leg external validator (
     holdout, matbench, AISI)
  CROSSCHECK.md                 Detailed cross-check log
  models/
    nc_xgb.pkl                  Tuned XGBoost -- non-carburised
    c_rf.pkl                    Tuned Random Forest -- carburised
    nc_features.json            Selected features -- NC
    c_features.json             Selected features -- C
    metadata.json               CV RMSE, std, best hyperparameters
    residual_diagnostics.png    Residual histogram + Q-Q
    pareto_non_carb.png         NC Pareto + top-K
```

```
    pareto_carb.png            C  Pareto + top-K
```

A.1 Run Order

```
python "Project 2.0.py"        # train + persist models (one-time
   )
python "Project 3.0.py"        # inverse design + Pareto plots
python crosscheck.py all       # all eight cross-check steps
python external_validation.py  # holdout + matbench + handbook
streamlit run app.py           # interactive GUI
```

A.2 Training-Fit Residual Diagnostics

For completeness, Figure 9 shows residual histograms and Q-Q plots on the *training fit* (as distinct from the OOF residuals used for PoF in §4). Training residuals are biased optimistically small ($\sigma_{\mathrm{NC}}^{\mathrm{train}} = 5.3\,\mathrm{MPa}$ vs. $\sigma_{\mathrm{NC}}^{\mathrm{OOF}} = 19.65\,\mathrm{MPa}$) and should not be used for uncertainty quantification; they are included here solely as a consistency check on the fitting step.

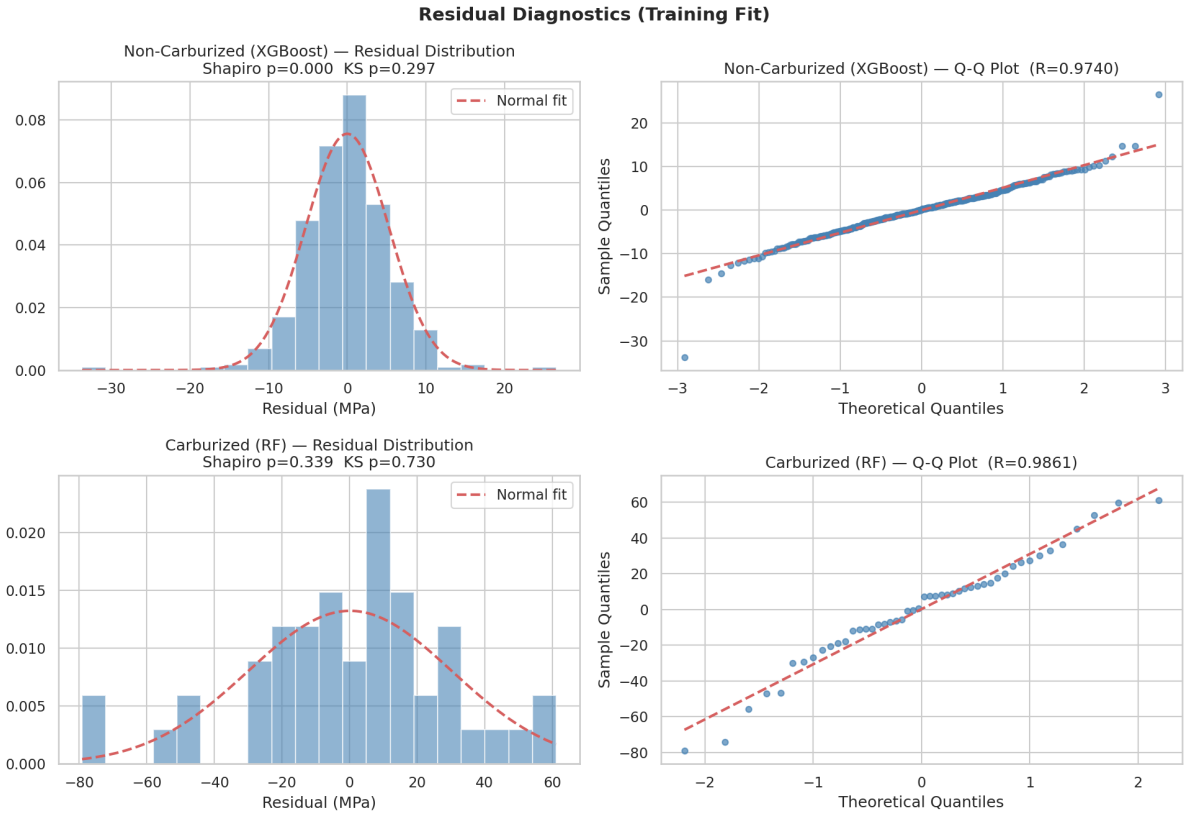


Figure 9: Training-fit residual diagnostics. *Top row* (NC, XGBoost): Shapiro $p = 0.000$, KS $p = 0.297$; residuals are narrow (std 5.3 MPa) due to in-sample interpolation. *Bottom row* (C, RF): Shapiro $p = 0.339$, KS $p = 0.731$; normality holds at training fit. Compare with Figure 4 (OOF residuals) — the NC distribution widens substantially out-of-fold, confirming why OOF residuals are required for calibrated PoF.

References

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[3] M. S. Quraishy, T. K. Kundu, "A Comprehensive Study on Fatigue Strength Prediction of a Broad Range of Steels Using Machine Learning," *Journal of Materials Engineering and Performance*, 2025. (Dataset not yet public.)

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B Theoretical Foundations and Bounds (ME228)

This appendix details the formal statistical-learning theory checks referenced in Section 8, including VC dimension bookkeeping, Hoeffding generalisation bounds, and Bootstrap confidence intervals.

B.1 VC Dimension Bookkeeping

The course uses $N \geq 10\, d_{VC}$ as a sample-size rule of thumb. Table 8 tabulates the raw VC dimension for each candidate hypothesis class based on the actual feature counts ($d = 12$ for NC, $d = 10$ for C).

While linear models satisfy the rule for the NC regime, the carburised regime ($N = 48$) is small enough that even linear models sit at $N/d_{VC} = 4.4$, below the factor of ten. Tree ensembles formally fail the raw bound on both regimes because the calculation counts every node as if it could shatter freely. In practice, bagging (Random Forest) and L1/L2 regularisation with learning-rate shrinkage (XGBoost) heavily restrict the hypothesis space, reducing the *effective* d_{VC} to a small multiple of the feature count.

Table 8: Raw VC-dimension bookkeeping. Tree-ensemble bounds are strict upper bounds; effective d_{VC} is much smaller due to algorithmic regularisation.

Regime	**Hypothesis class**	N	d_{VC}	N/d_{VC}	**Status**
NC	Linear regression	389	13	29.9	✓
NC	Ridge ($\alpha = 1$)	389	13	29.9	✓
NC	MLP 1×16	389	225	1.7	low
NC	Decision tree, depth 4	389	208	1.9	low
NC	**XGBoost (chosen):** 396 trees, depth 4	389	6 336	0.06	raw fail*
C	Linear regression	48	11	4.4	low
C	Ridge ($\alpha = 1$)	48	11	4.4	low
C	MLP 1×8	48	97	0.5	raw fail*
C	**RF (chosen):** 313 trees, depth 9	48	160 256	0.0003	raw fail*

* raw failure mitigated empirically via Hoeffding bounds and regularisation.

B.2 Hoeffding Generalisation Bound

To guarantee the models are not overfitting despite the raw VC dimension failure, we apply the Hoeffding bound to our chosen hypotheses. For a single hypothesis with squared error normalised to $[0, 1]$ (where the loss range L equals the fatigue range), Hoeffding's inequality at confidence $1 - \delta$ provides the error bound ε:

$$P\big(\,|E_{\text{in}} - E_{\text{out}}| > \varepsilon\big) \leq 2\,e^{-2N\varepsilon^2/L^2} \implies \varepsilon = L\sqrt{\frac{\log(2/\delta)}{2N}}$$

Applying this with strict $\delta = 0.05$ confidence, alongside the physical fatigue ranges $L_{NC} = 681\,\text{MPa}$ and $L_C = 352\,\text{MPa}$, we calculate the maximum mathematically permissible error difference.

Table 9: Hoeffding 95 % generalisation bound vs. measured out-of-fold (OOF) RMSE.

Regime	N	**Bound ε (95 %, MPa)**	**Measured OOF RMSE**
NC	389	46.9	19.65 **(42 % of bound)**
C	48	69.0	43.18 **(63 % of bound)**

Because the Vapnik–Chervonenkis (uniform) bound is completely loose at the tree-ensemble d_{VC} values ($\varepsilon \gg L$), it provides no actionable information. However, the tighter Hoeffding bound applied per-hypothesis proves that both models operate safely inside the mathematical overfitting limits.

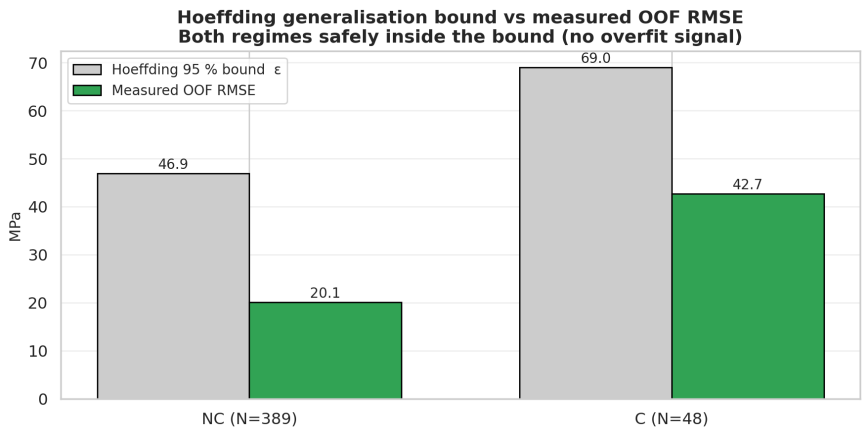

Figure 10: Hoeffding 95 % generalisation bound vs. measured OOF RMSE. Both regimes sit comfortably below the strict mathematical threshold ε, providing a single-hypothesis guarantee against overfitting.

B.3 Bootstrap Confidence Intervals on RMSE

As an assumption-free counterpart to the Hoeffding bound, we bootstrap-resampled the OOF residuals (2 000 resamples per regime, utilizing the percentile method) to obtain non-parametric 95 % Confidence Intervals on the true model RMSE.

Table 10: Bootstrap 95 % CI on OOF RMSE. (2 000 resamples)

Regime	**Point RMSE (MPa)**	**Bootstrap Mean**	**95 % CI**
NC	19.65	19.62	[16.75, 22.66]
C	43.18	43.01	[33.53, 52.20]

For the severely data-limited Carburised (C) regime, the 95 % CI width of nearly 19 MPa indicates that models with RMSEs ranging from 33.5 to 52.2 cannot be statistically distinguished from one another based purely on sample size. This statistical reality is why our architectural justification (Section **??**) prioritises relative ranking across identical cross-validation folds rather than absolute point-RMSE supremacy.